SQL Execution Plan -

Hi friends today we will learn how sql make execution plan visually explained them.

It is an exactly pin point tool to find why your query is slow

Large query have lot of joins and aggregation so it is slow
So to improve performance we used indexes

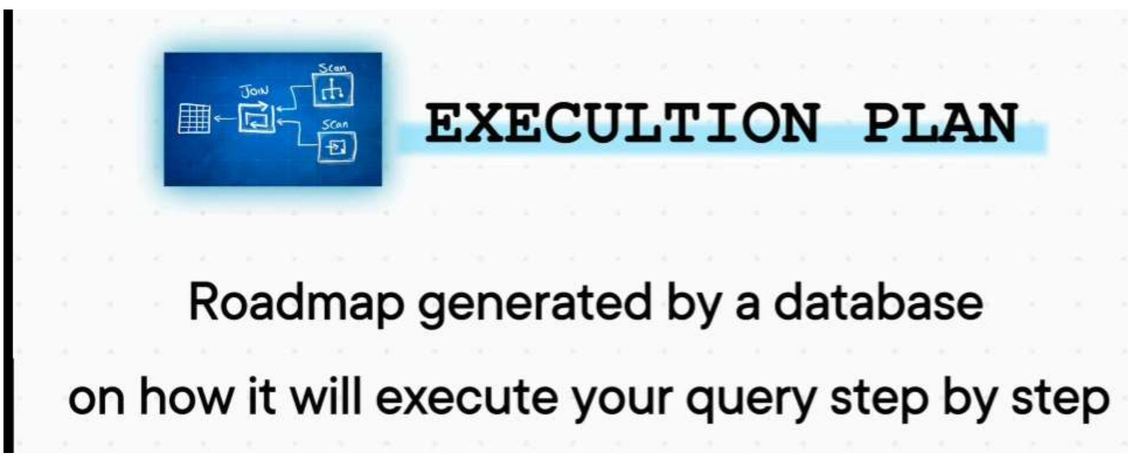
Where exactly we are going build the indexes in which table in which columns
So for this we have to understand where exactly is the problem is it in sorting the data or joining tables or in aggregating data or by sorting data.

Now in order to answer there questions we have execution plan

How the data base exactly process your query step by step
It shows where is exactly performance issue.

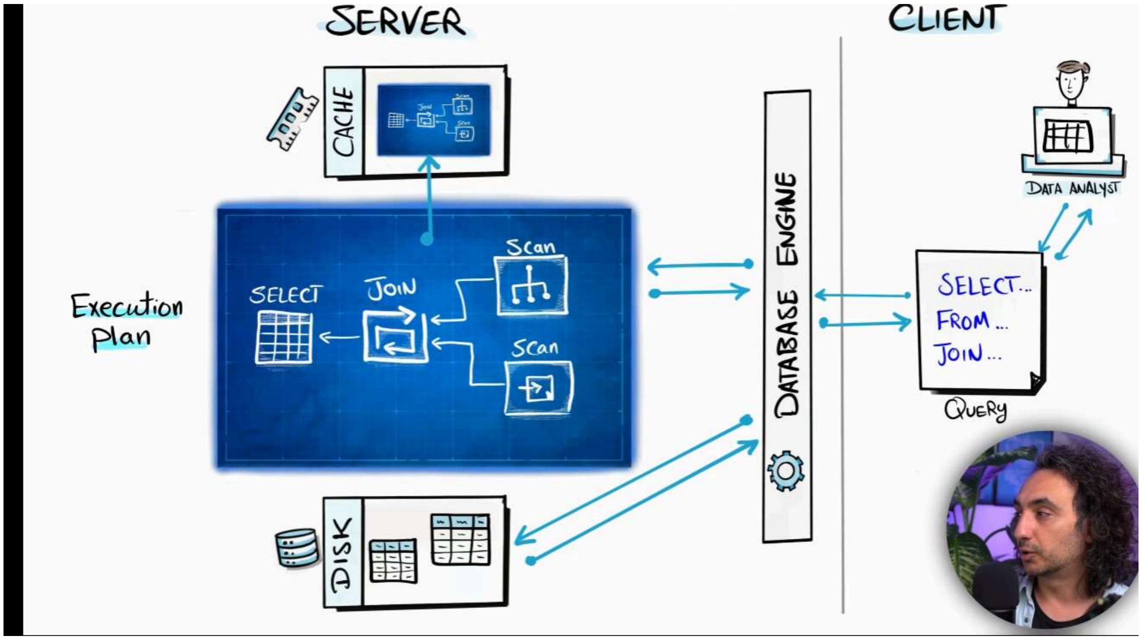
So, execution plan is like a window to understand how the sql db exactly thinks.
And once we understand then we can make a right decision fro building an index to improve performance or solving issues.

```sql
JOIN DimSalesChannel pc ON s.SalesChannelKey = pc.SalesChannelKey
GROUP BY pc.SalesChannel, YEAR(s.OrderDate), DATEPART(QUARTER, s.OrderDate);

-- Query 6: Customer demographics and purchase behavior by region
SELECT c.CustomerKey, c.Gender, c.MaritalStatus, st.SalesTerritoryRegion,
       COUNT(s.SalesOrderNumber) AS OrderCount, SUM(s.SalesAmount) AS TotalSpent
FROM FactInternetSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
JOIN DimSalesTerritory st ON s.SalesTerritoryKey = st.SalesTerritoryKey
GROUP BY c.CustomerKey, c.Gender, c.MaritalStatus, st.SalesTerritoryRegion;

-- Query 7: Revenue by product category and year for the last decade
SELECT pc.ProductCategoryKey, YEAR(s.OrderDate) AS SalesYear,
       SUM(s.SalesAmount) AS AnnualRevenue
FROM FactInternetSales s
JOIN DimProductCategory pc ON s.ProductCategoryKey = pc.ProductCategoryKey
WHERE s.OrderDateKey BETWEEN 20100101 AND 20201231
GROUP BY pc.ProductCategoryKey, YEAR(s.OrderDate);

-- Query 8: Monthly trend analysis by customer segment and reseller
SELECT c.CustomerSegment, r.ResellerKey, MONTH(s.OrderDate) AS SalesMonth,
       SUM(s.SalesAmount) AS MonthlyRevenue
FROM FactResellerSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
JOIN DimReseller r ON s.ResellerKey = r.ResellerKey
GROUP BY c.CustomerSegment, r.ResellerKey, MONTH(s.OrderDate);

-- Query 9: Product sales performance in each territory
SELECT st.SalesTerritoryKey, st.SalesTerritoryCountry, ps.ProductSubcategoryKey,
       COUNT(s.SalesOrderNumber) AS SalesCount, SUM(s.SalesAmount) AS SalesTotal
FROM FactResellerSales s
JOIN DimSalesTerritory st ON s.SalesTerritoryKey = st.SalesTerritoryKey
JOIN DimProductSubcategory ps ON s.ProductSubcategoryKey = ps.ProductSubcategoryKey
GROUP BY st.SalesTerritoryKey, st.SalesTerritoryCountry, ps.ProductSubcategoryKey;

-- Query 10: Quarterly sales trends for high-value customers
SELECT c.CustomerKey, DATEPART(QUARTER, s.OrderDate) AS SalesQuarter,
       SUM(s.SalesAmount) AS QuarterlySales
FROM FactInternetSales s
JOIN DimCustomer c ON s.CustomerKey = c.CustomerKey
WHERE s.SalesAmount > 1000
GROUP BY c.CustomerKey, DATEPART(QUARTER, s.OrderDate);

-- Query 11: Employee sales performance by quarter and year
```

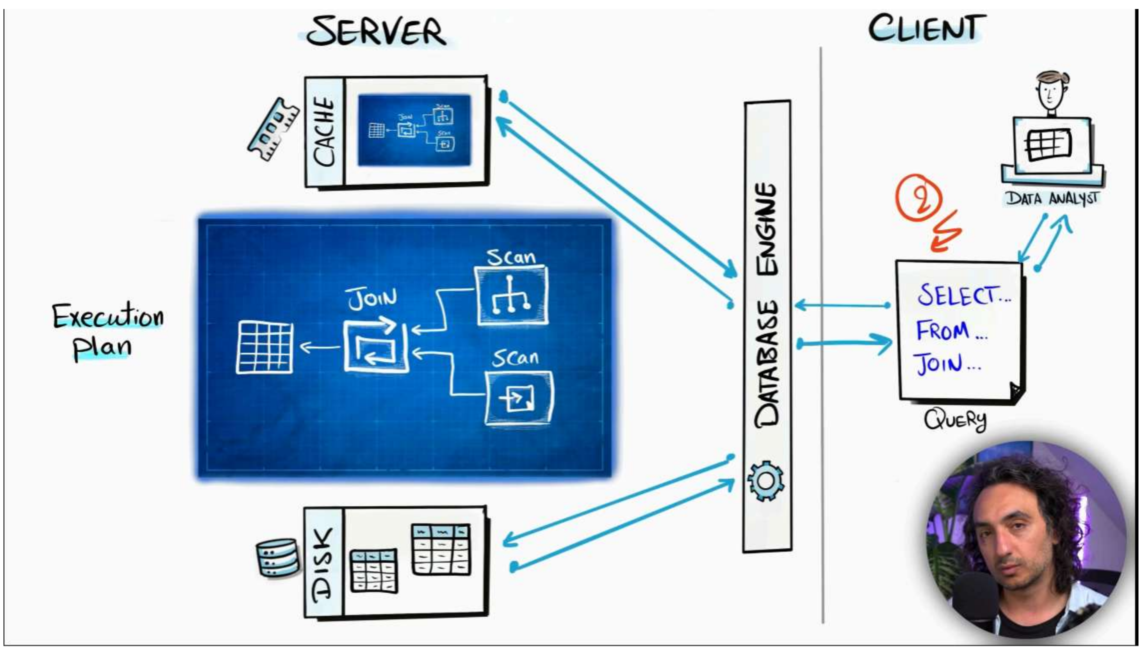


Execution plan like finding roadmap over map to reach at destination.
It check how to scan table like how to get data from tables like to get multiple tables.

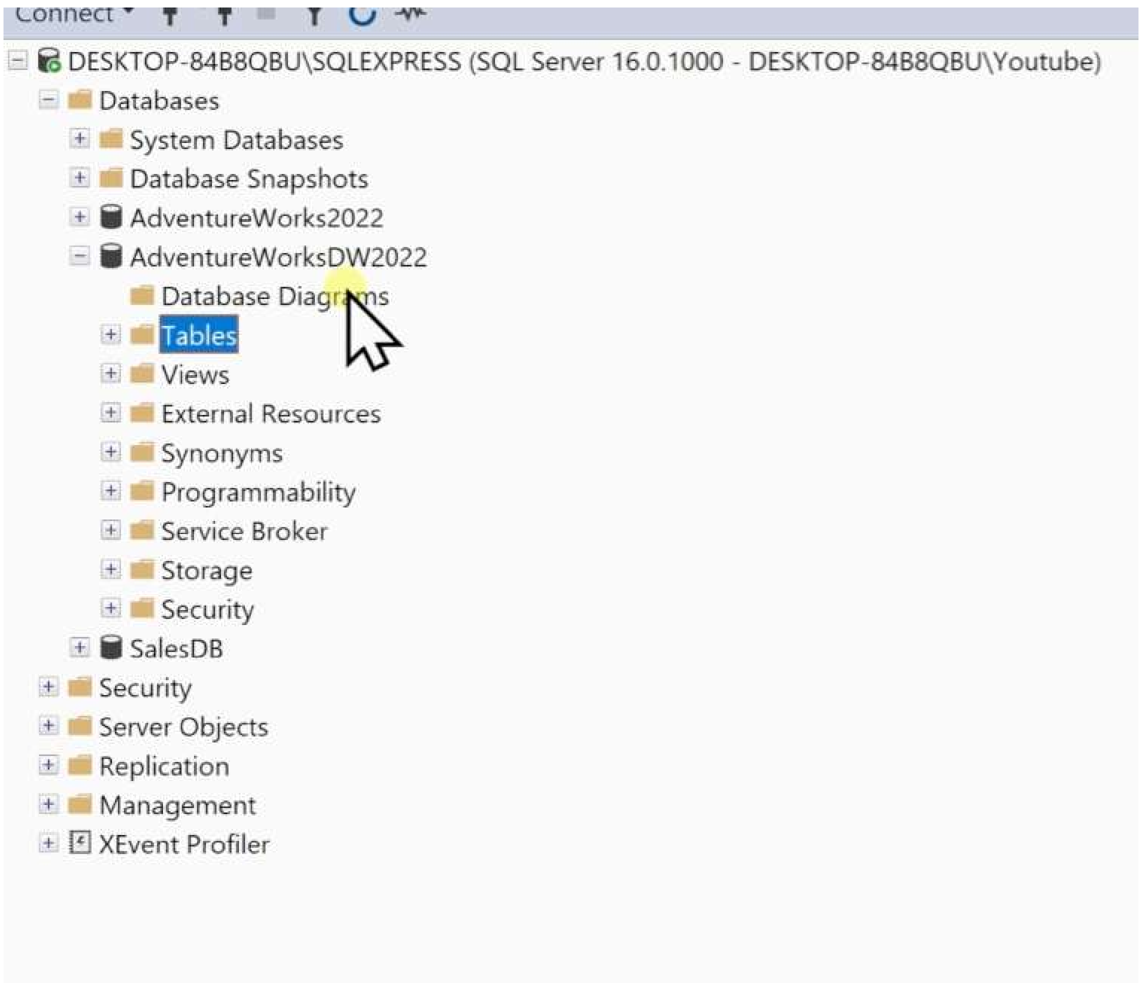
To decide it's full table scan or index scan then decide which type of join loop join single join then select statement.

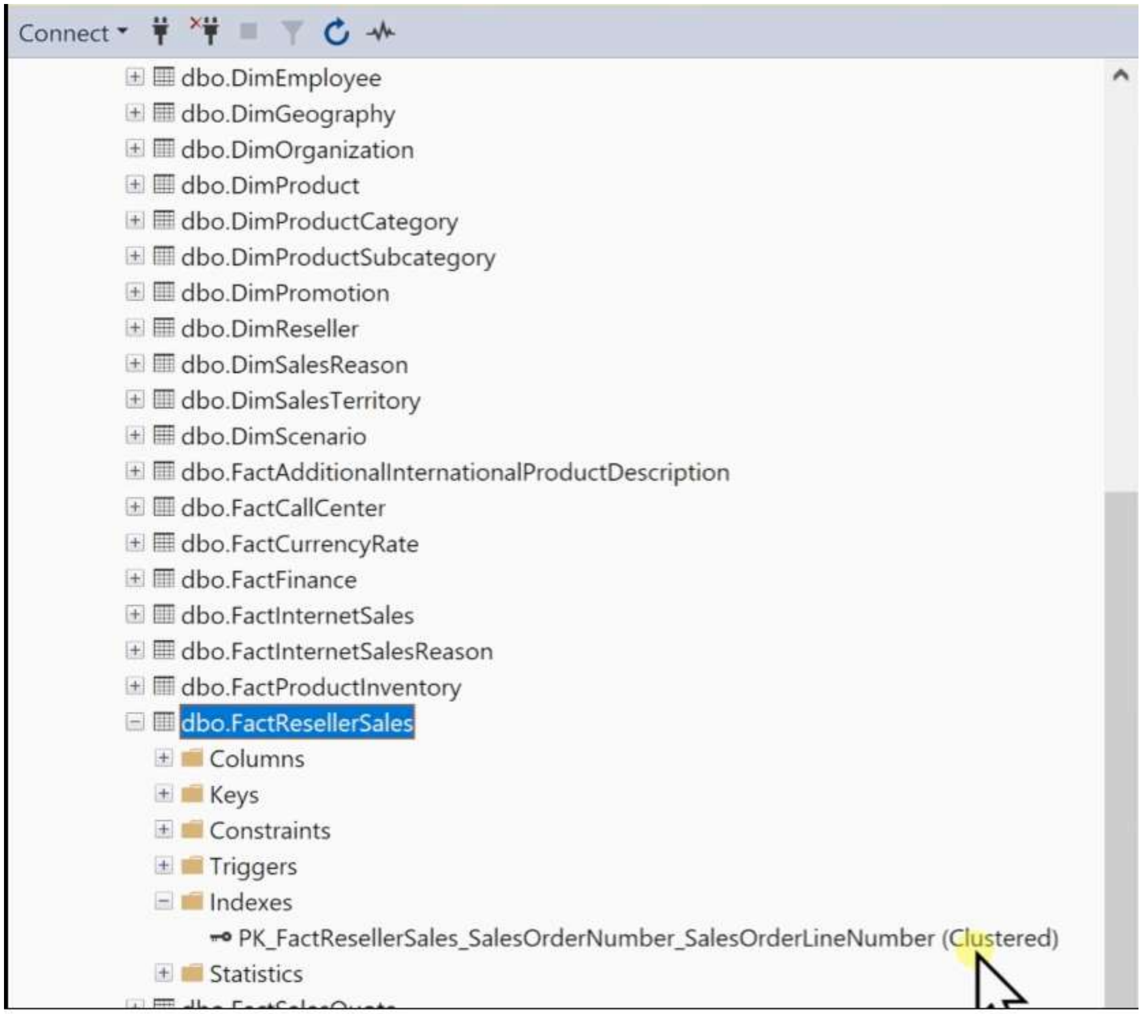


When you run same query again - takes help from cache store execution plan

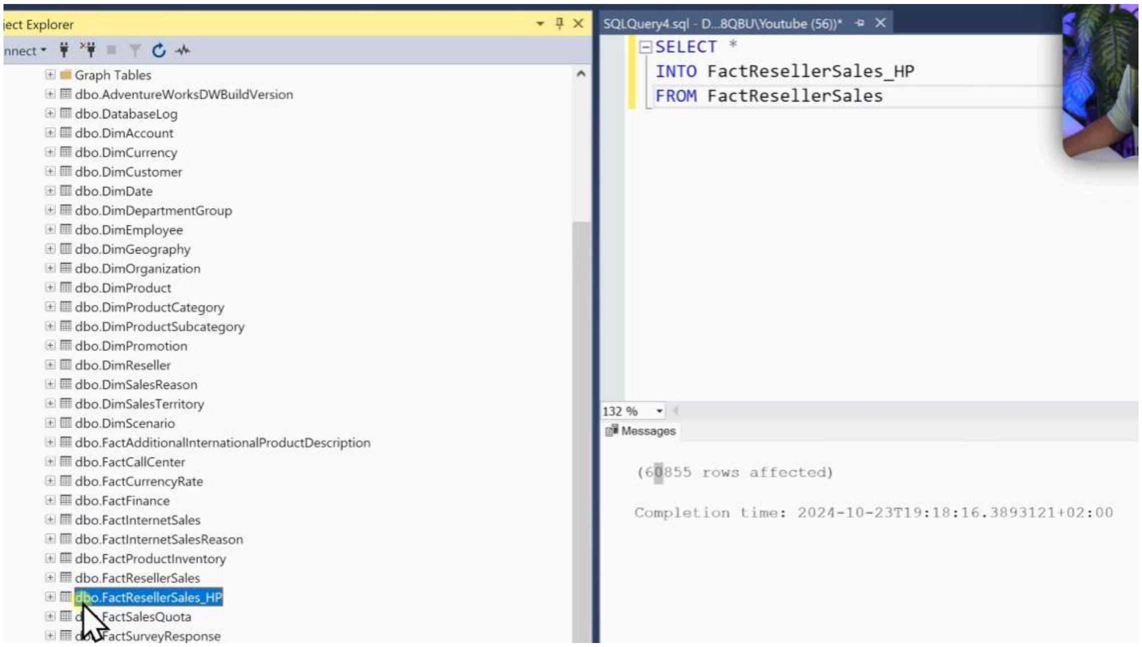


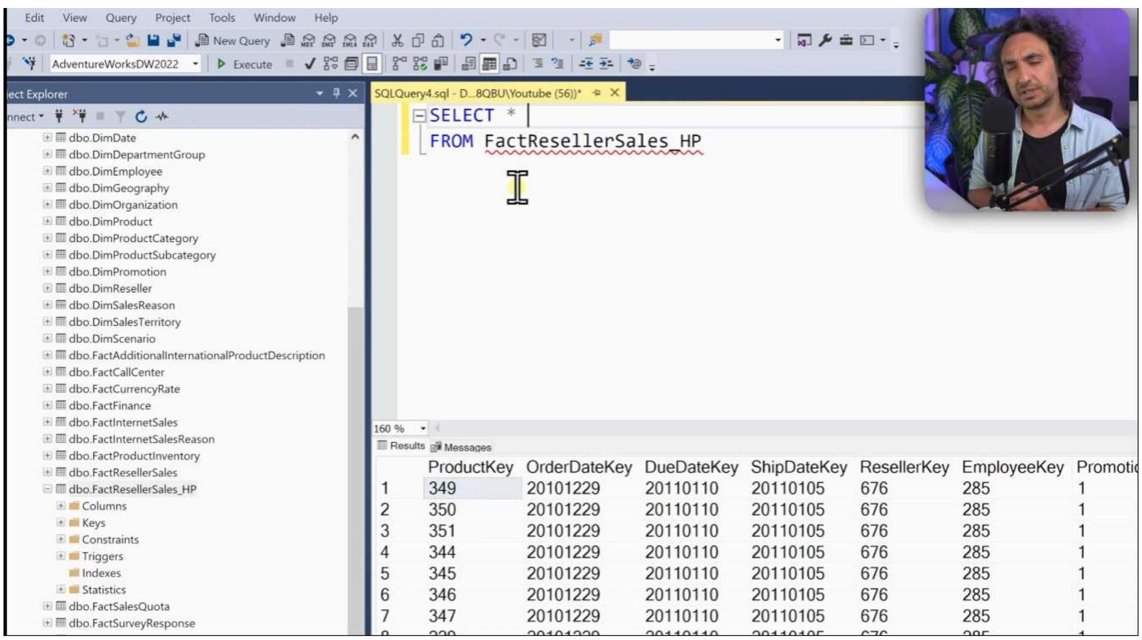

Execution Plan
Basics


Connect
DESKTOP-84B8QBU\SQLEXPRESS (SQL Server 16.0.1000 - DESKTOP-84B8QBU\Youtube)
Databases
System Databases
Database Snapshots
AdventureWorks2022
AdventureWorksDW2022
Database Diagrams
Tables
Views
External Resources
Synonyms
Programmability
Service Broker
Storage
Security
SalesDB
Security
Server Objects
Replication
Management
XEvent Profiler



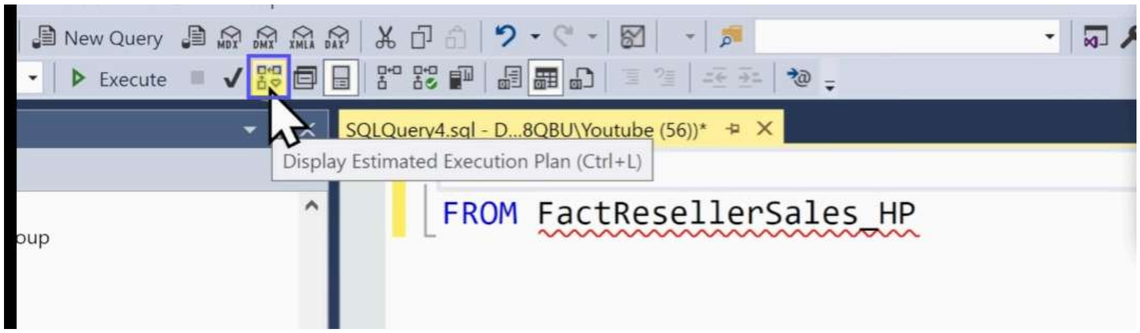
Create same mirror table I mean creating same table without index

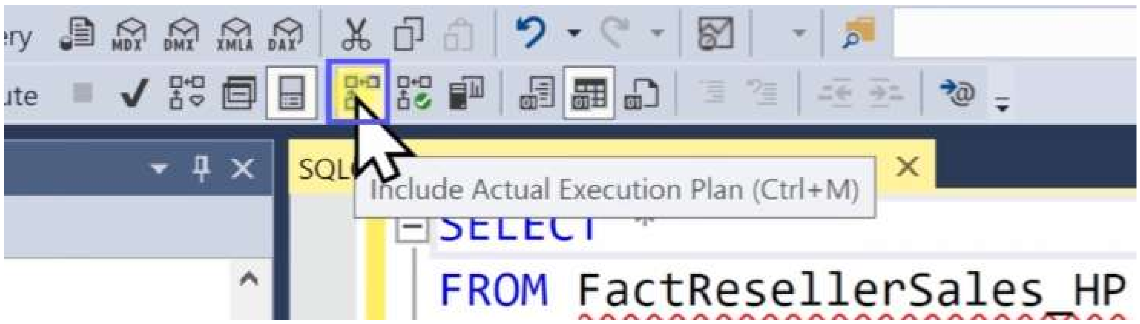


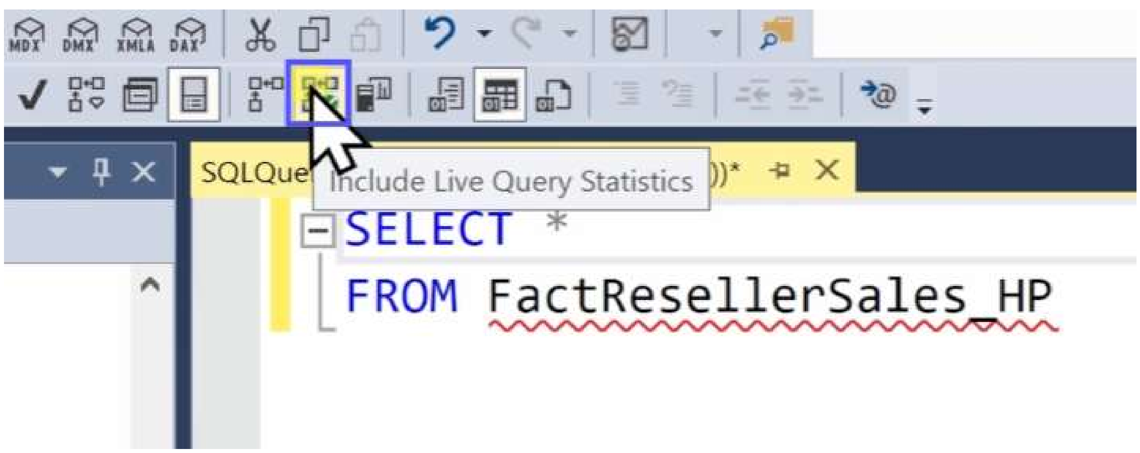


How to check execution plan of this query.

Go to at top tool bar

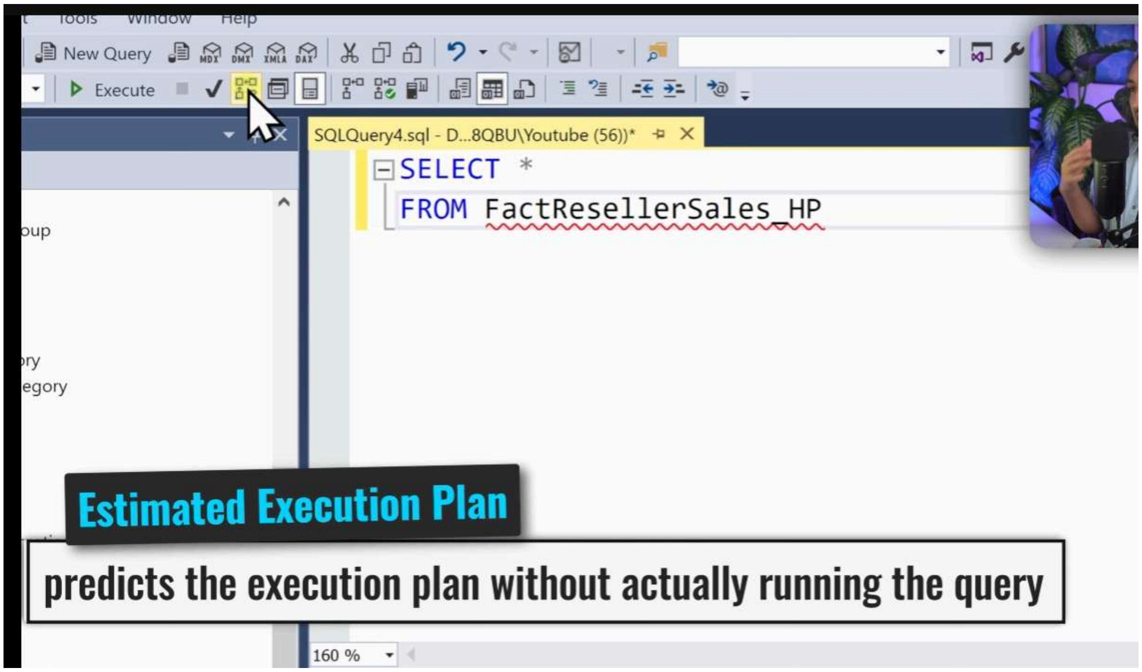
1. 

2. 

3. 

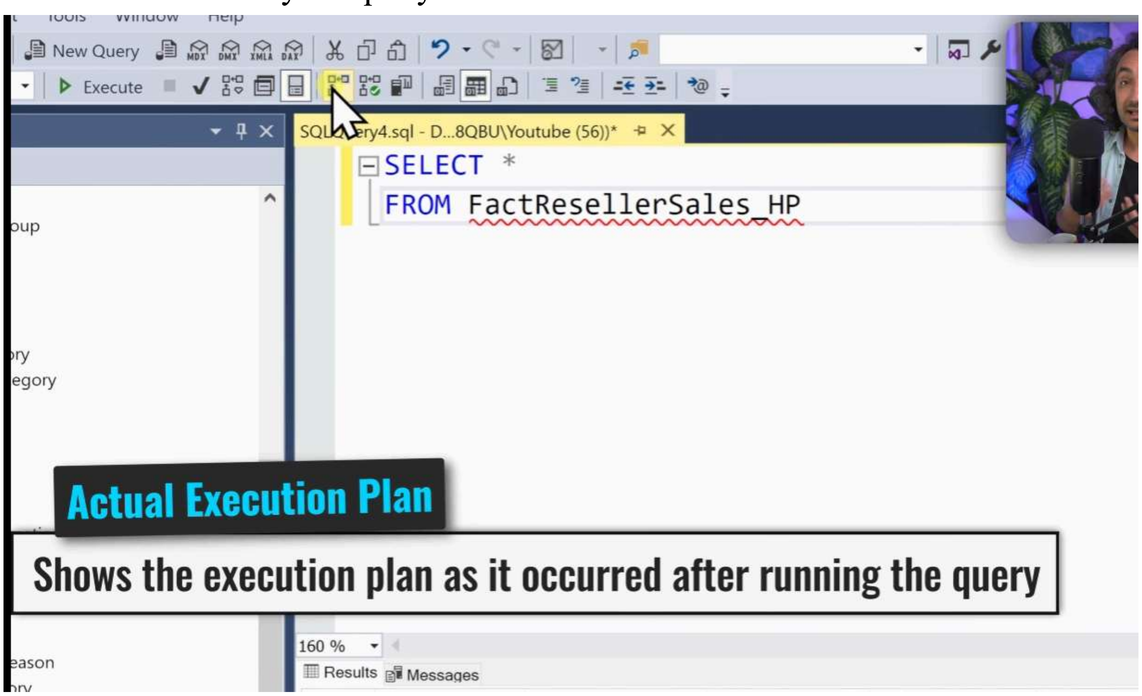
Now check what are difference between them let’s go with first one.
First one is guess only without execution the query

It is before execution of your query.



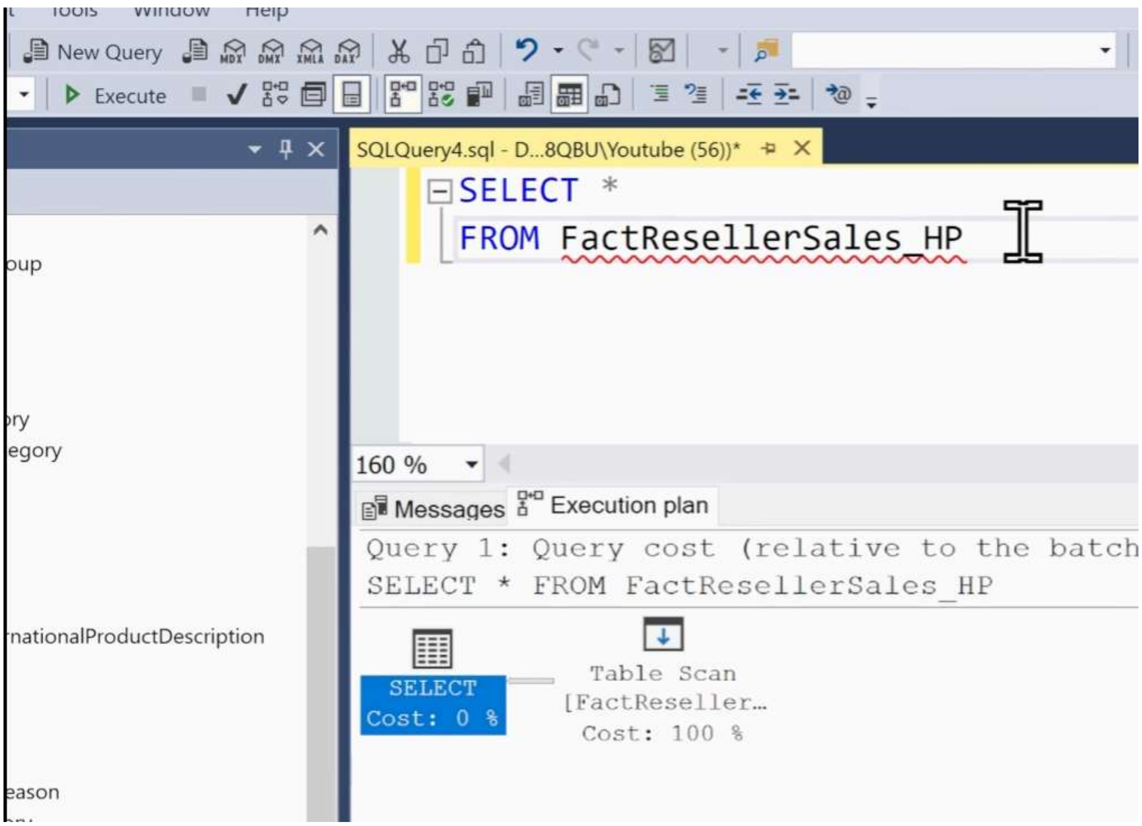
Second one is actual one show execution plan that show you that is used to process your query

It is after execution your query

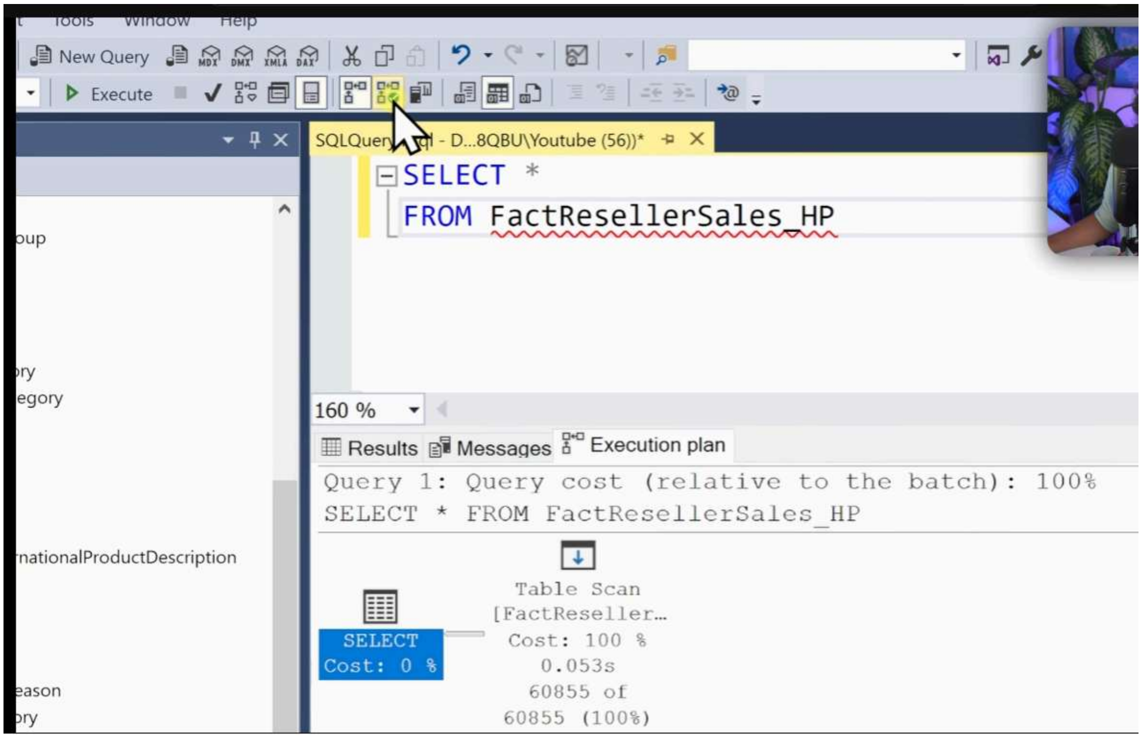


Third one is live execution plan - it is while execution of your query.

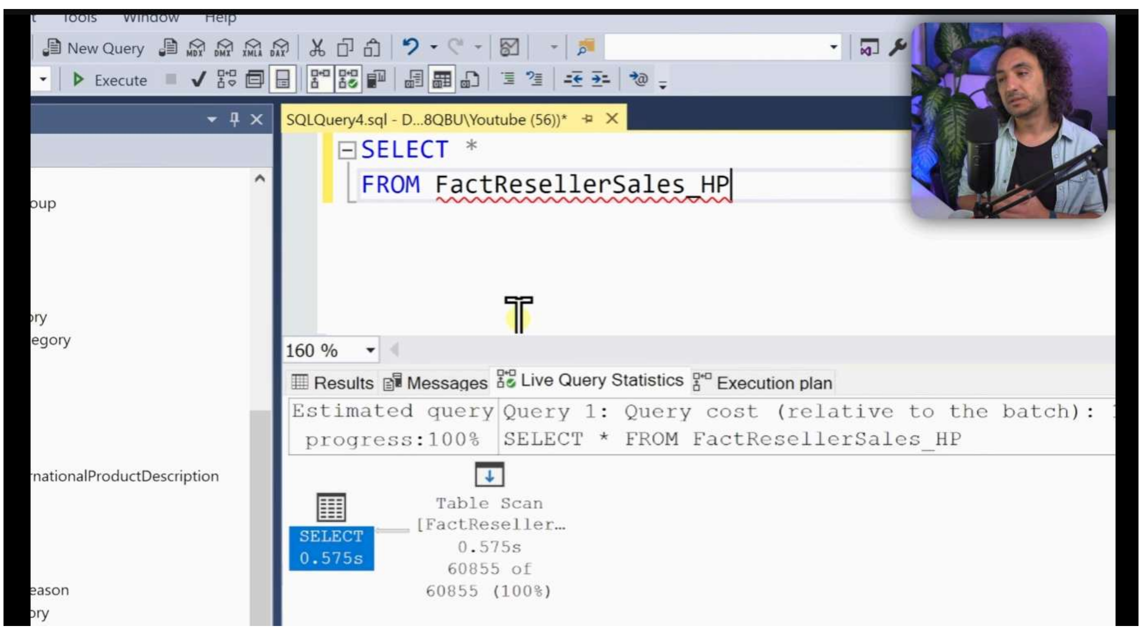
First one

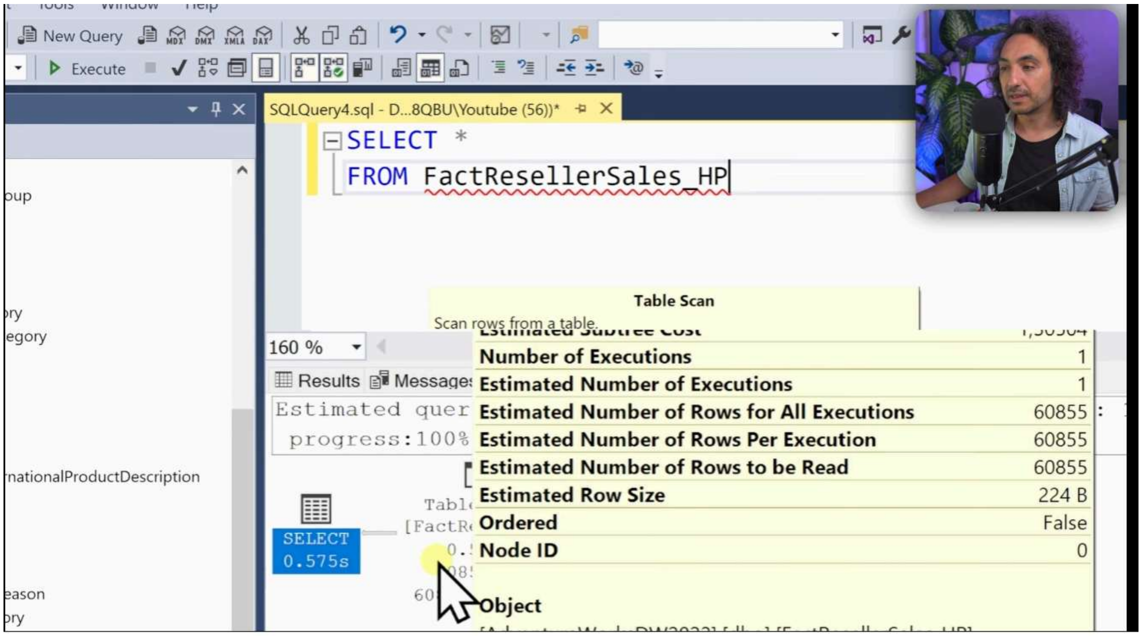


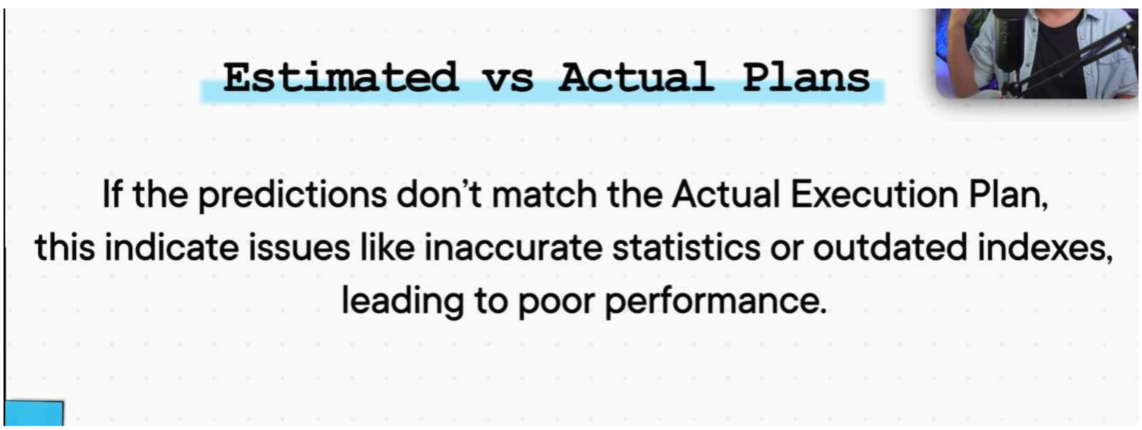
Second one


New Query
Execute
SQLQuery4.sql - D...8QBU\Youtube (56))*
SELECT *
FROM FactResellerSales_HP
160 %
Results
Messages
Execution plan
Query 1: Query cost (relative to the batch): 100%
SELECT * FROM FactResellerSales_HP
Table Scan
[FactReseller...
Cost: 100 %
0.053s
60855 of
60855 (100%)
SELECT
Cost: 0 %

Third one

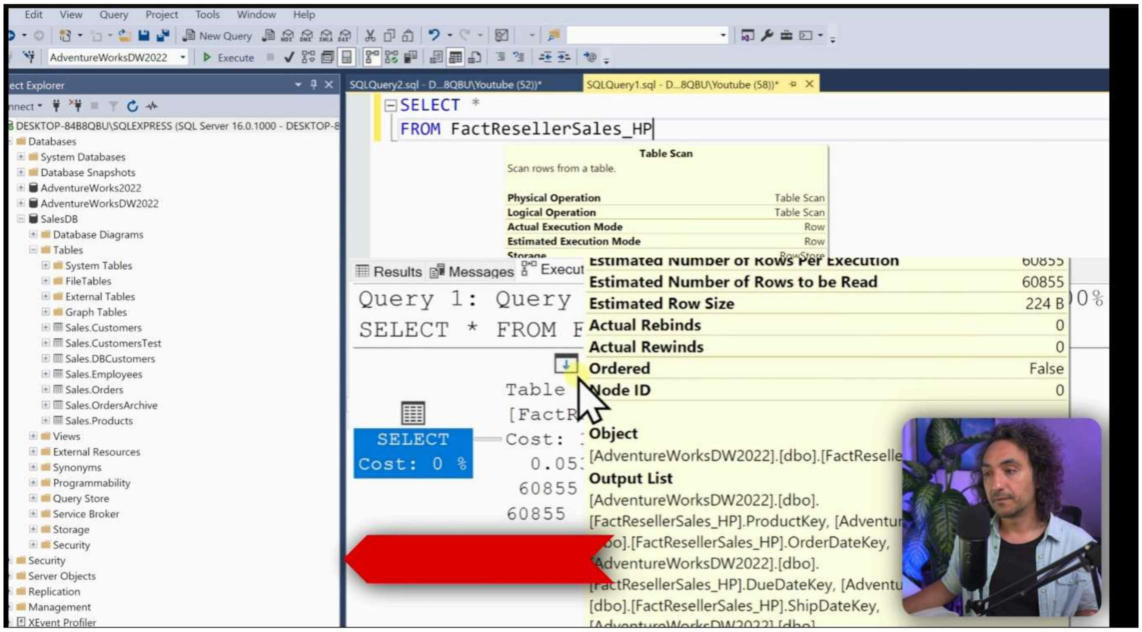

New Query
Execute
SQLQuery4.sql - D...8QBU\Youtube (56))*
SELECT *
FROM FactResellerSales_HP
160 %
Results
Messages
Live Query Statistics
Execution plan
Estimated query progress:100%
Query 1: Query cost (relative to the batch):
SELECT * FROM FactResellerSales_HP
Table Scan
[FactReseller...
0.575s
60855 of
60855 (100%)
SELECT
0.575s





Now check execution plan for clustered index and heap.
Actual execution plan

Heap execution plan - reads from right to left


SELECT *
FROM FactResellerSales_HP
Table Scan
Scan rows from a table.
Physical Operation Table Scan
Logical Operation Table Scan
Actual Execution Mode Row
Estimated Execution Mode Row
Estimated Number of Rows to be Read 60855
Estimated Row Size 224 B
Actual Rebinds 0
Actual Rewinds 0
Ordered False
Node ID 0
Object
Output List
SELECT
Cost: 0 %

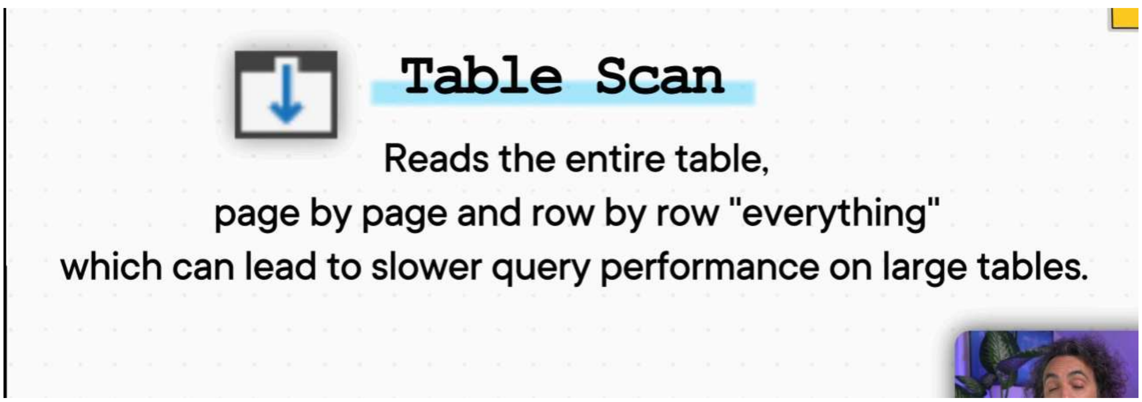
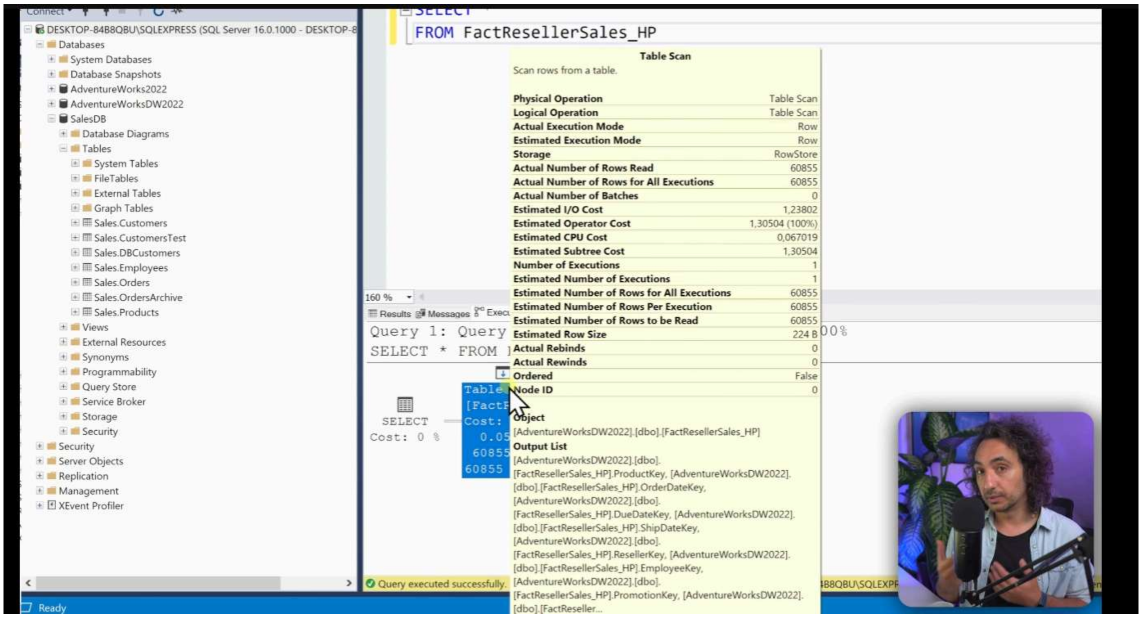
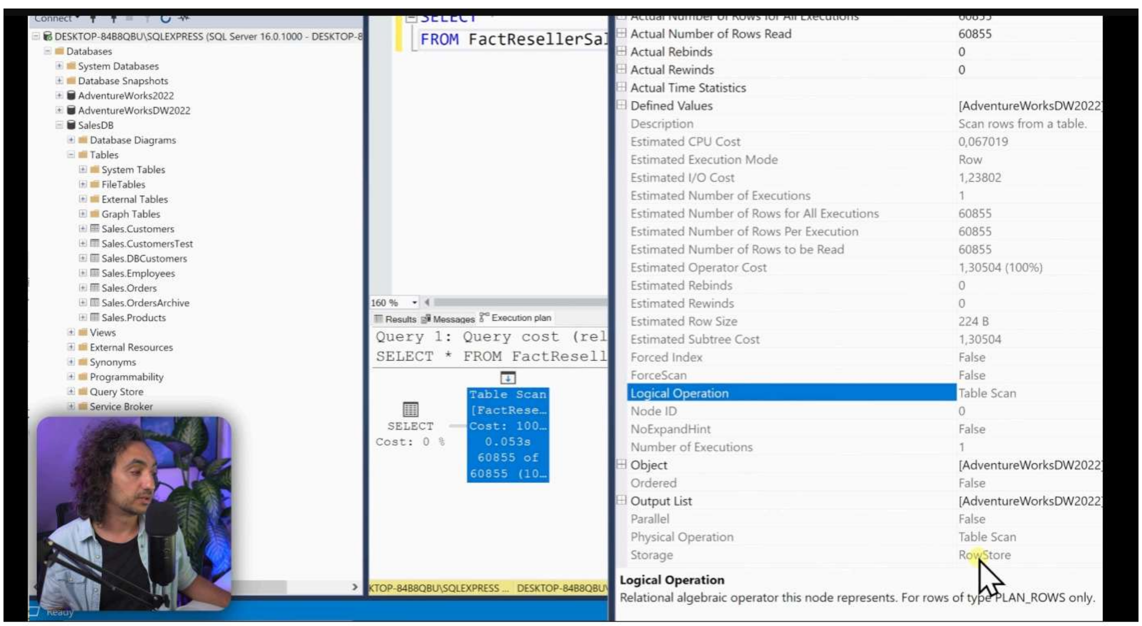


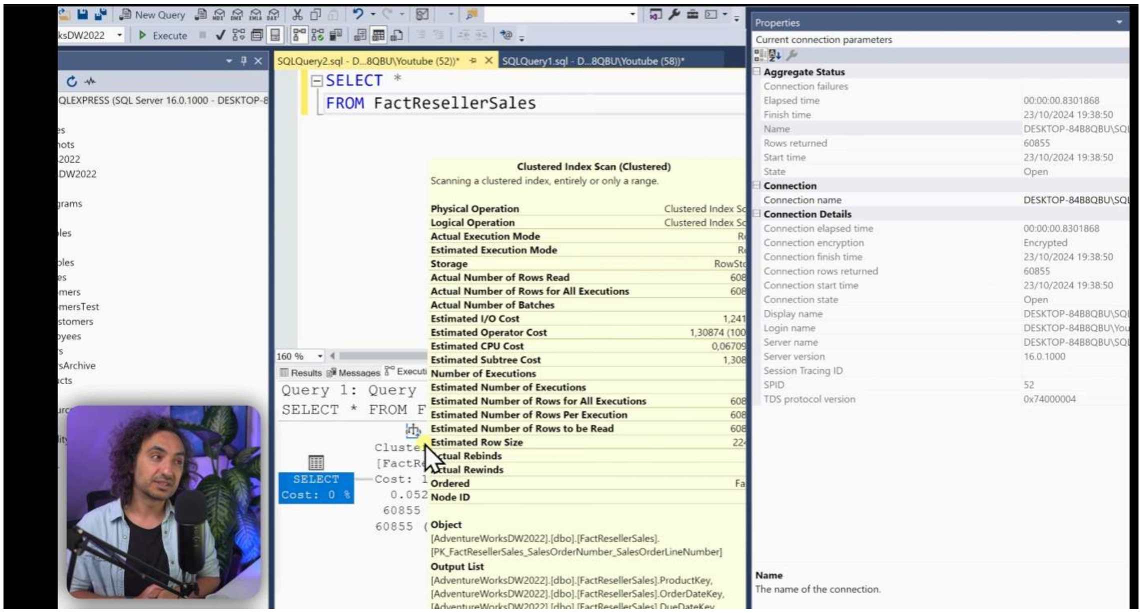
Table Scan
Reads the entire table,
page by page and row by row "everything"
which can lead to slower query performance on large tables.



Check by right click on table and got to properties read would be easy.

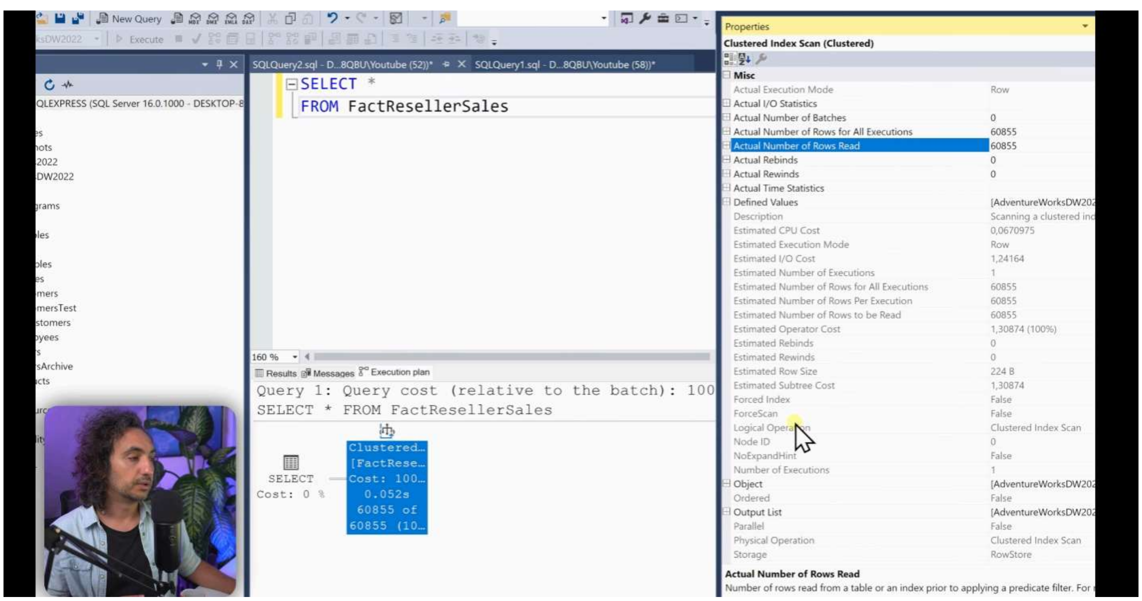


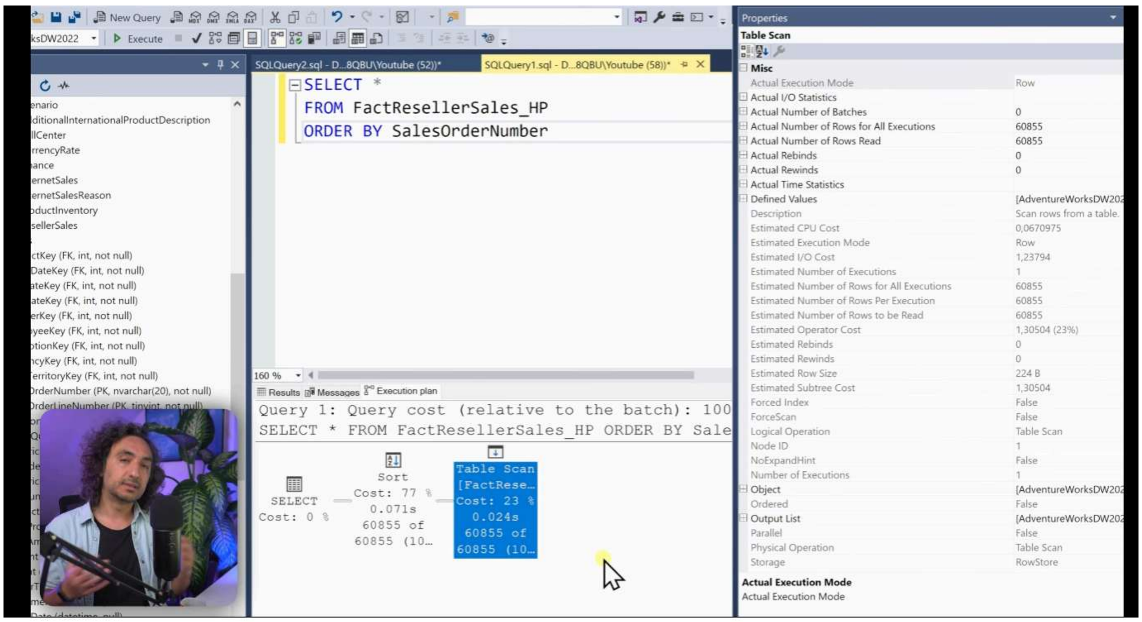
Clustered index execution plan

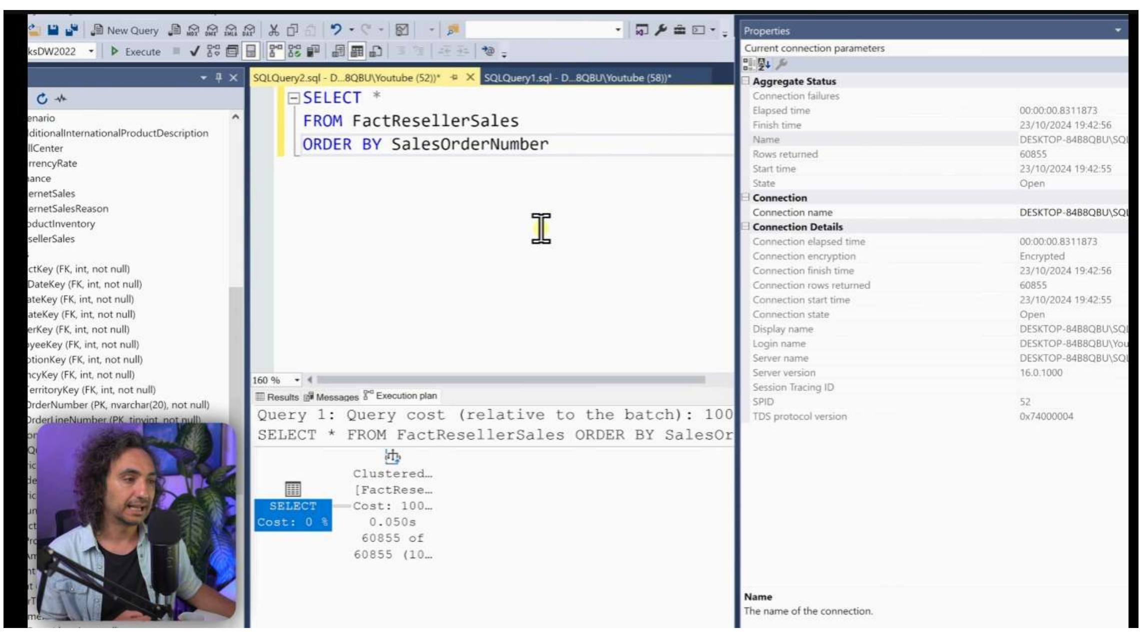


Index Scan

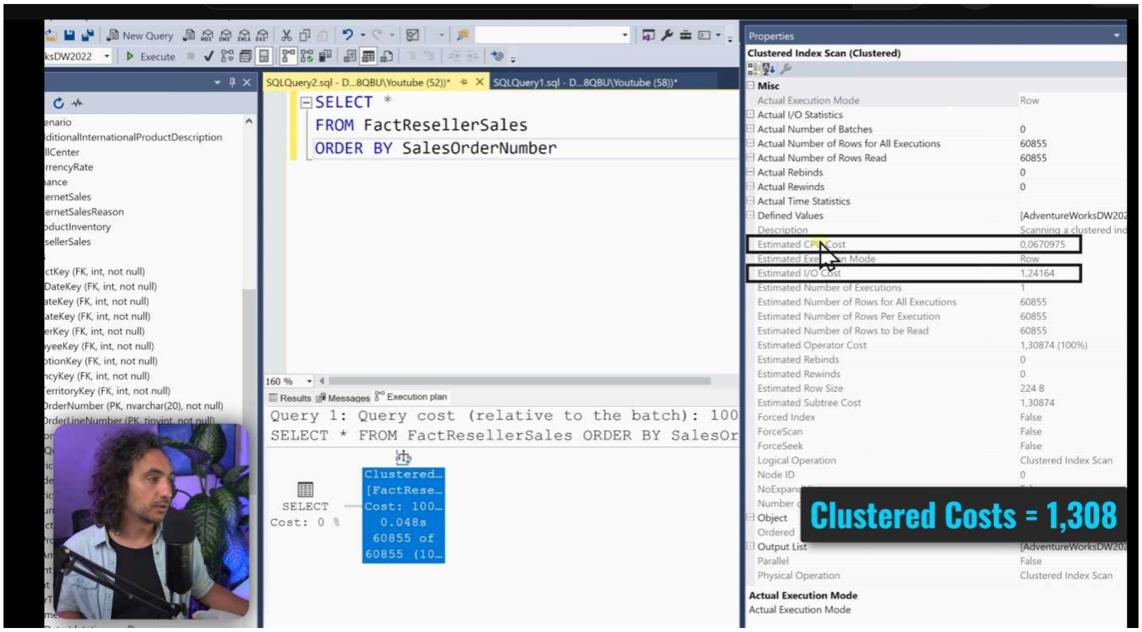
Scans all data in an index to find matching rows.

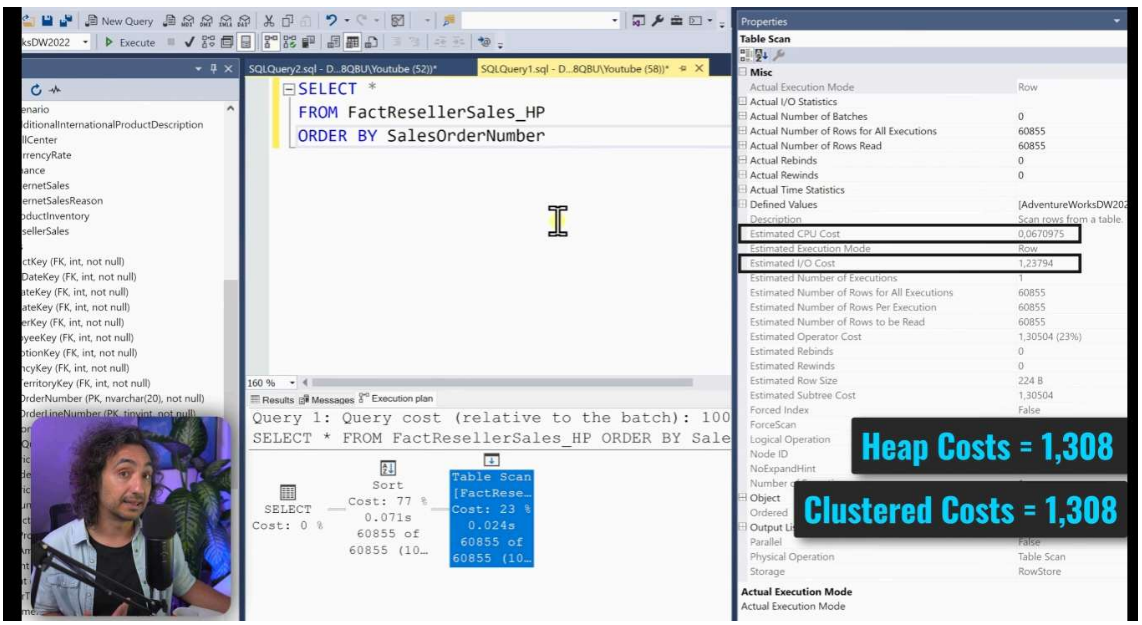


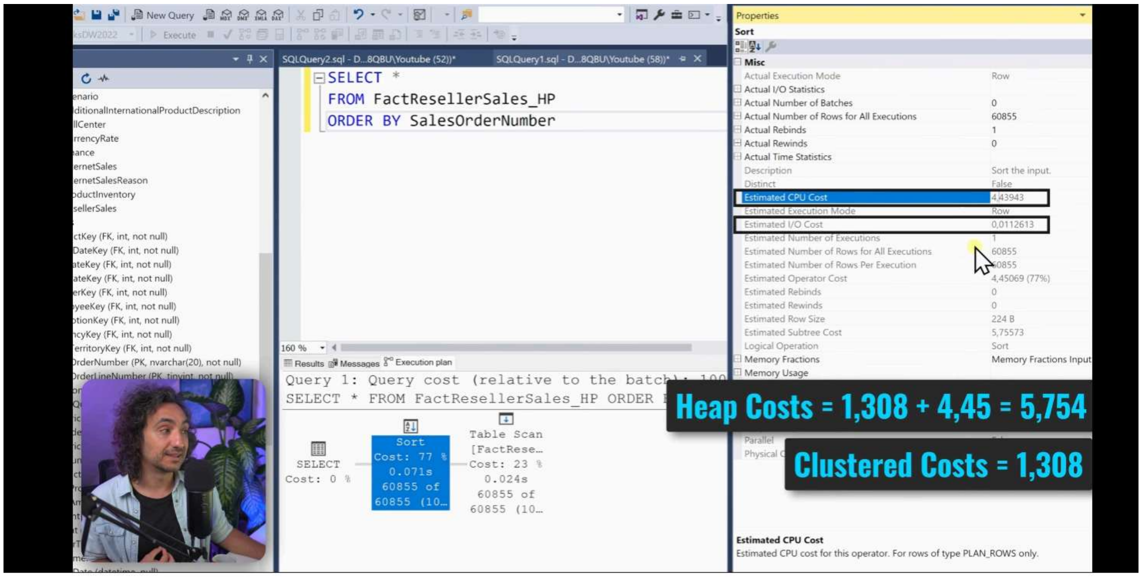




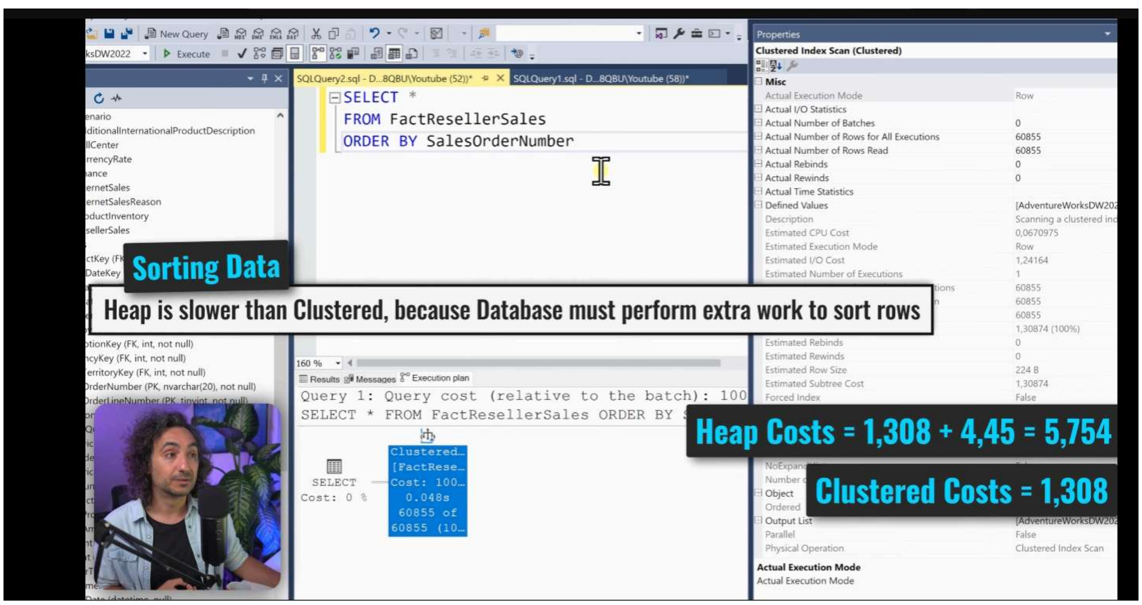
No extra step to do sort in clustered index because data is already sorted.

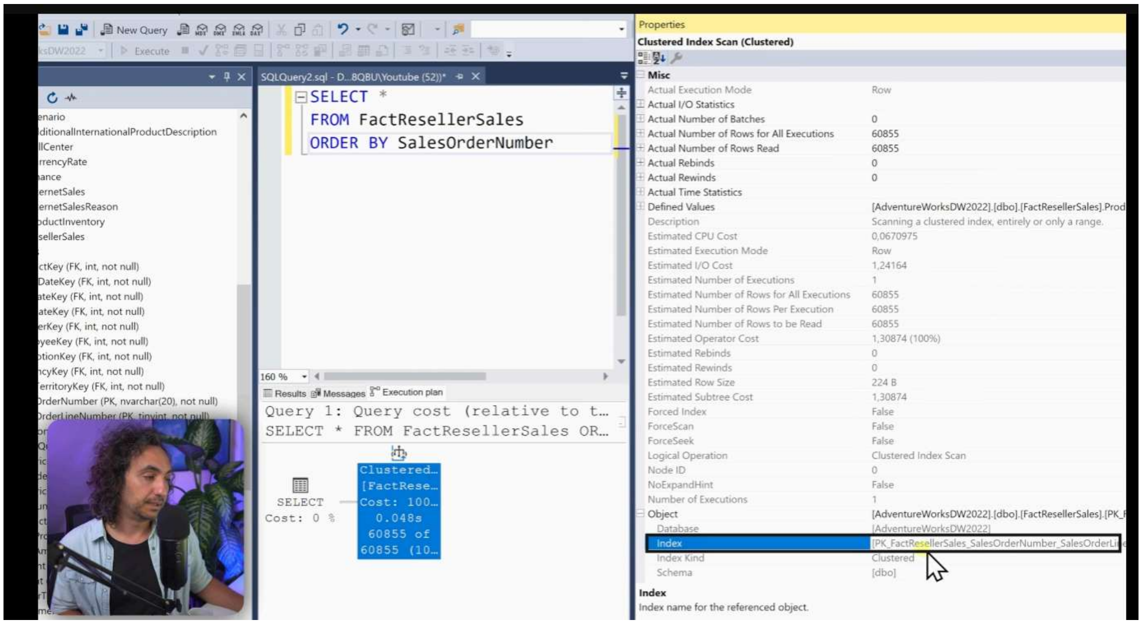

SELECT *
FROM FactResellerSales
ORDER BY SalesOrderNumber
Query 1: Query cost (relative to the batch): 100
SELECT * FROM FactResellerSales ORDER BY SalesOr
Clustered Index Scan (Clustered)
Estimated CPU Cost 0,0670975
Estimated I/O Cost 1,24164
Estimated Operator Cost 1,30874 (100%)
Estimated Subtree Cost 1,30874
Clustered Costs = 1,308

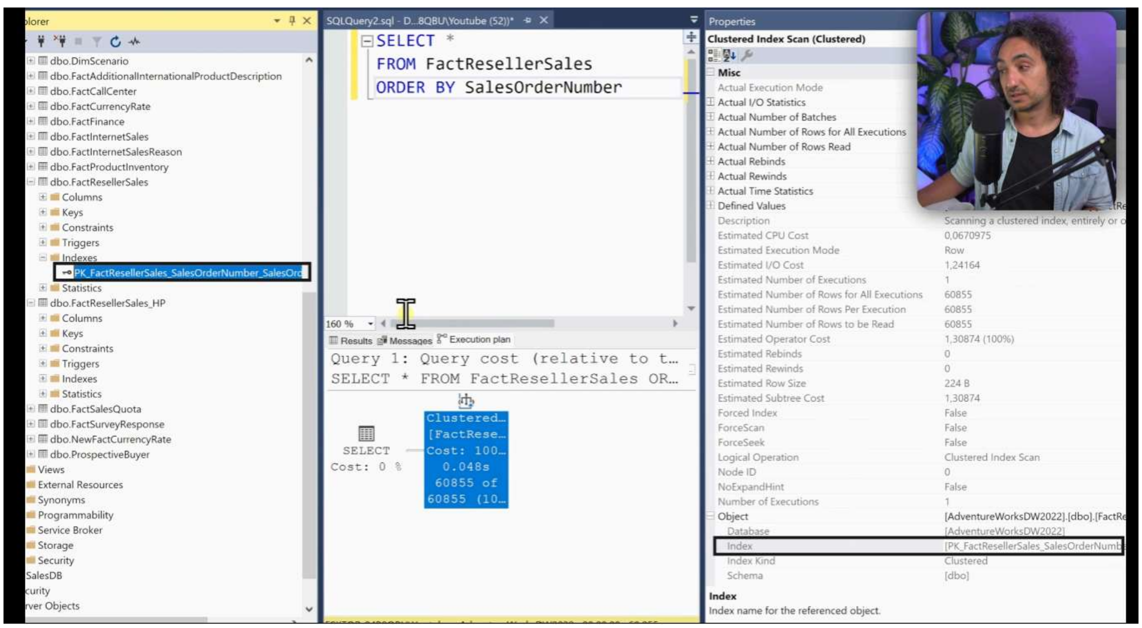

SELECT *
FROM FactResellerSales_HP
ORDER BY SalesOrderNumber
Query 1: Query cost (relative to the batch): 100
SELECT * FROM FactResellerSales_HP ORDER BY Sale
Table Scan
Estimated CPU Cost 0,0670975
Estimated I/O Cost 1,23794
Estimated Operator Cost 1,30504 (23%)
Estimated Subtree Cost 1,30504
Heap Costs = 1,308
Clustered Costs = 1,308



High cost for sorting the data.





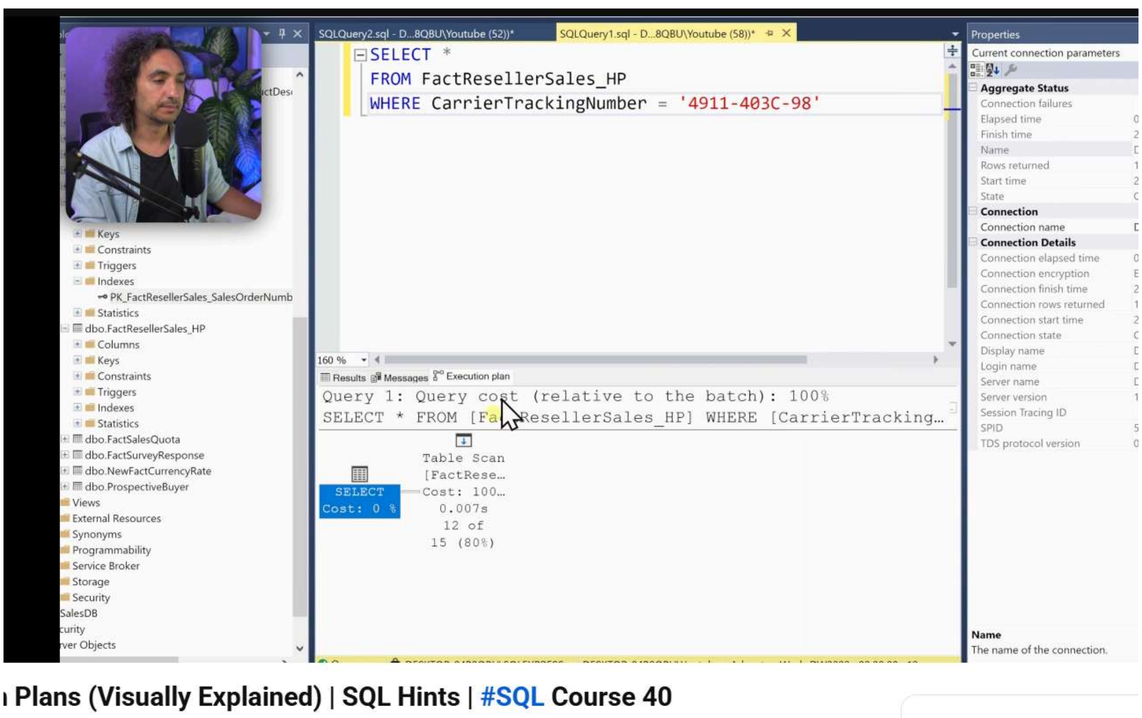


In execution plan you can find/check it which index is you are using in your query.

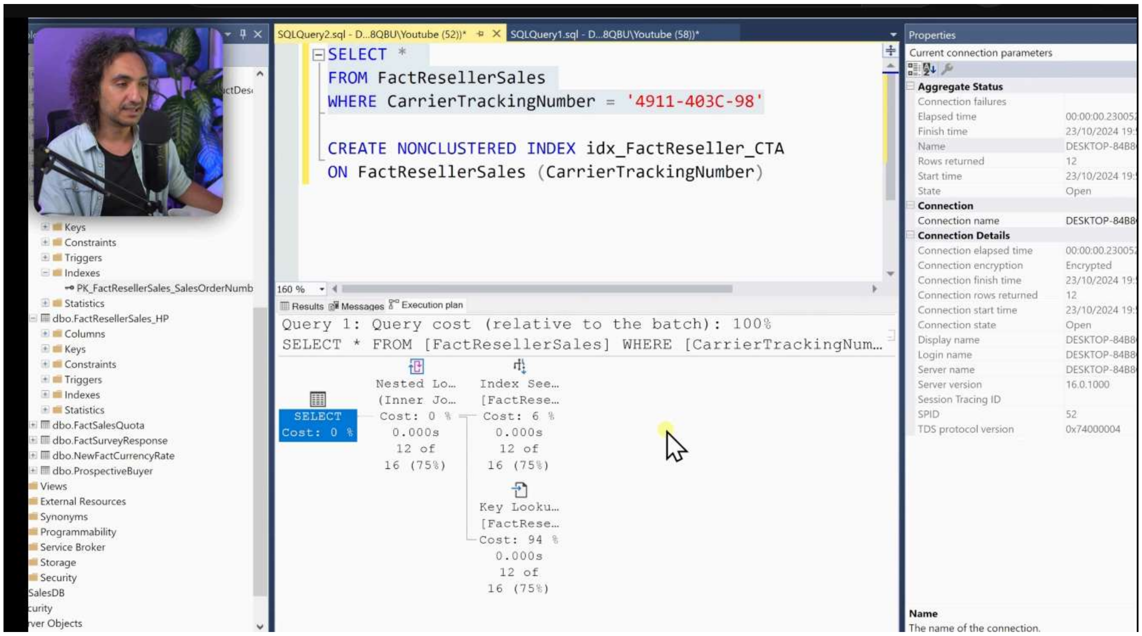
And this is important if you create new index and run query and check db is using your newly created index or not if not then this is wrong decisions about your index.
So each time your create your index and check through execution plan

Instead of pk we used any one of those column which we are using in our table.

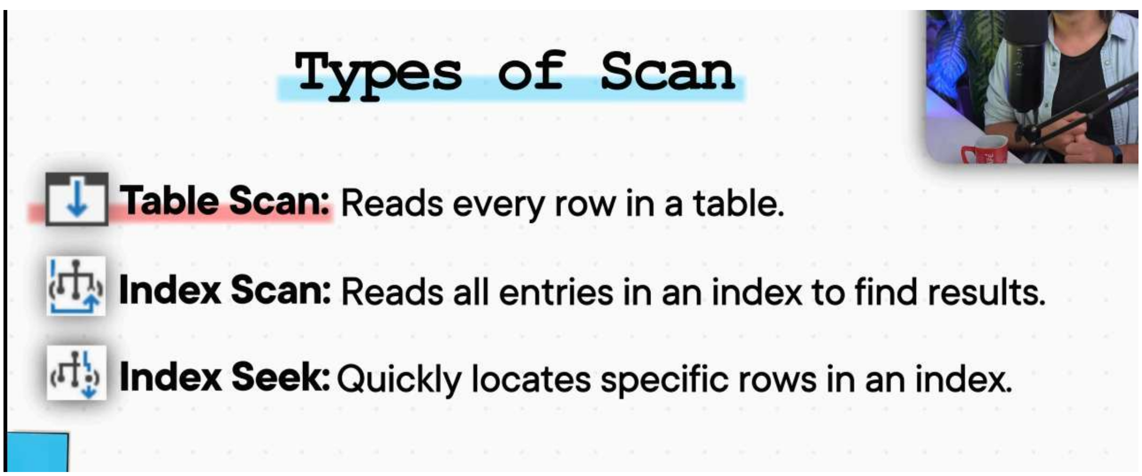
Ex- carrier tracking number



We have table scan in execution plan



Now we have index seek in above



So , there are three types of scan best one is index seek and worst one is table scan

In index sacn no of rows to be executed is 12 we are reading 12 rows only in order to get 12
Cost is very less and it's very fast